

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

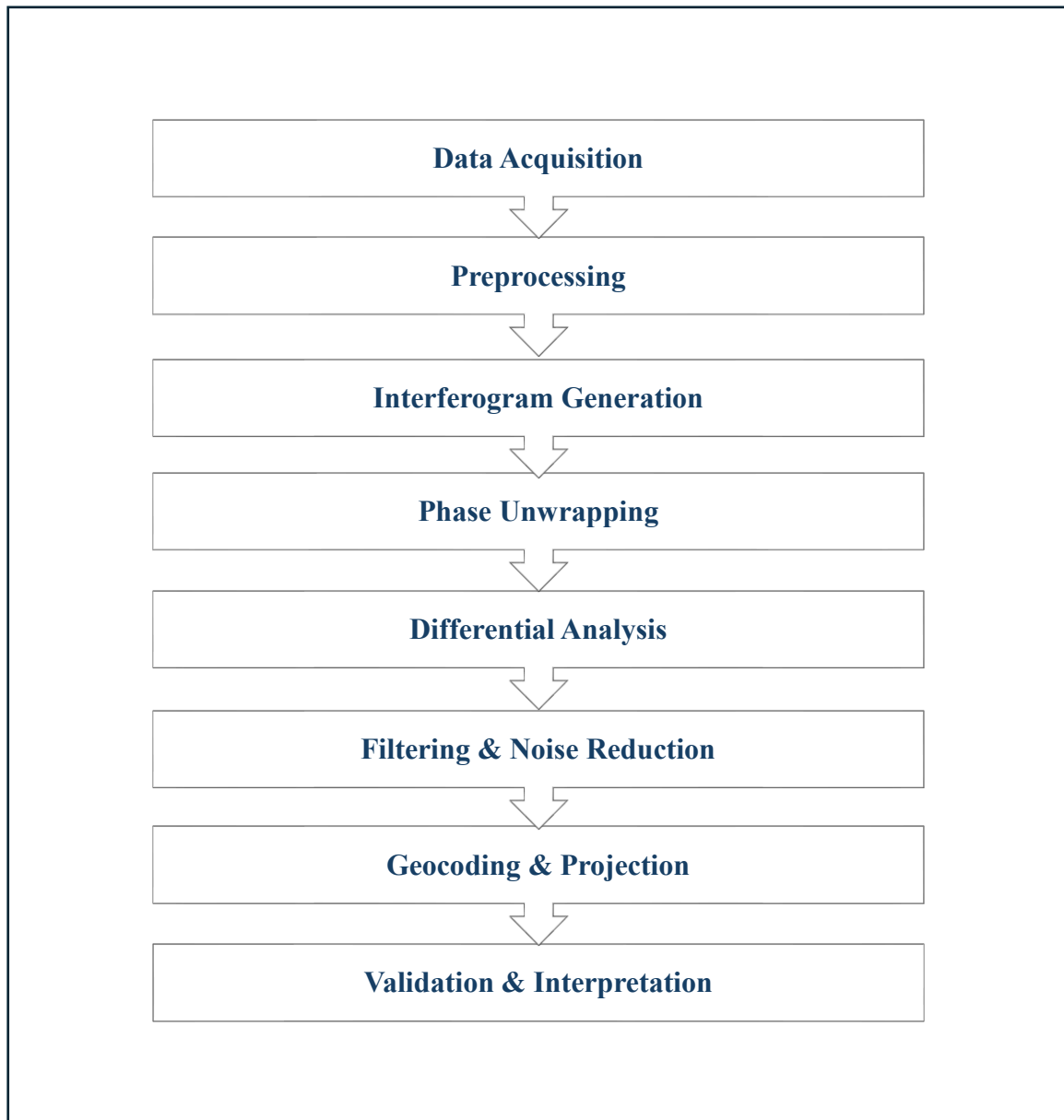
Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 920+200

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

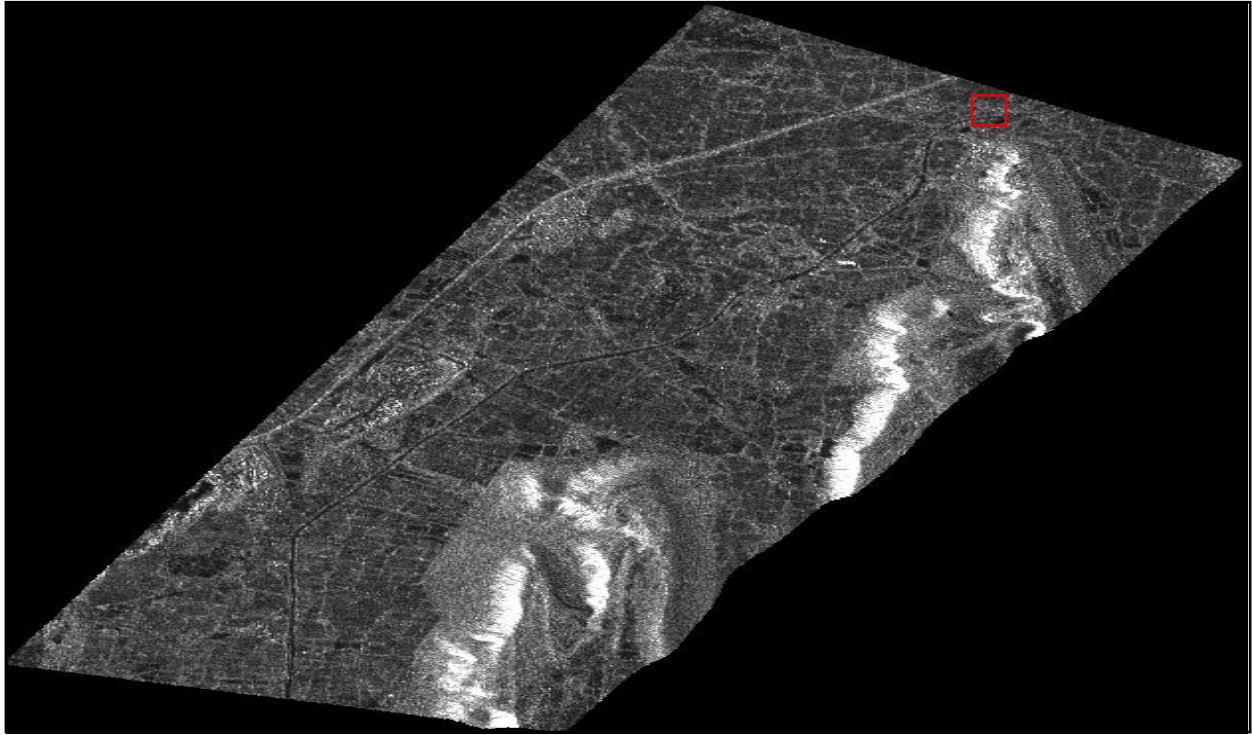
Flight Direction - DESCENDING

Polarization – VV+VH

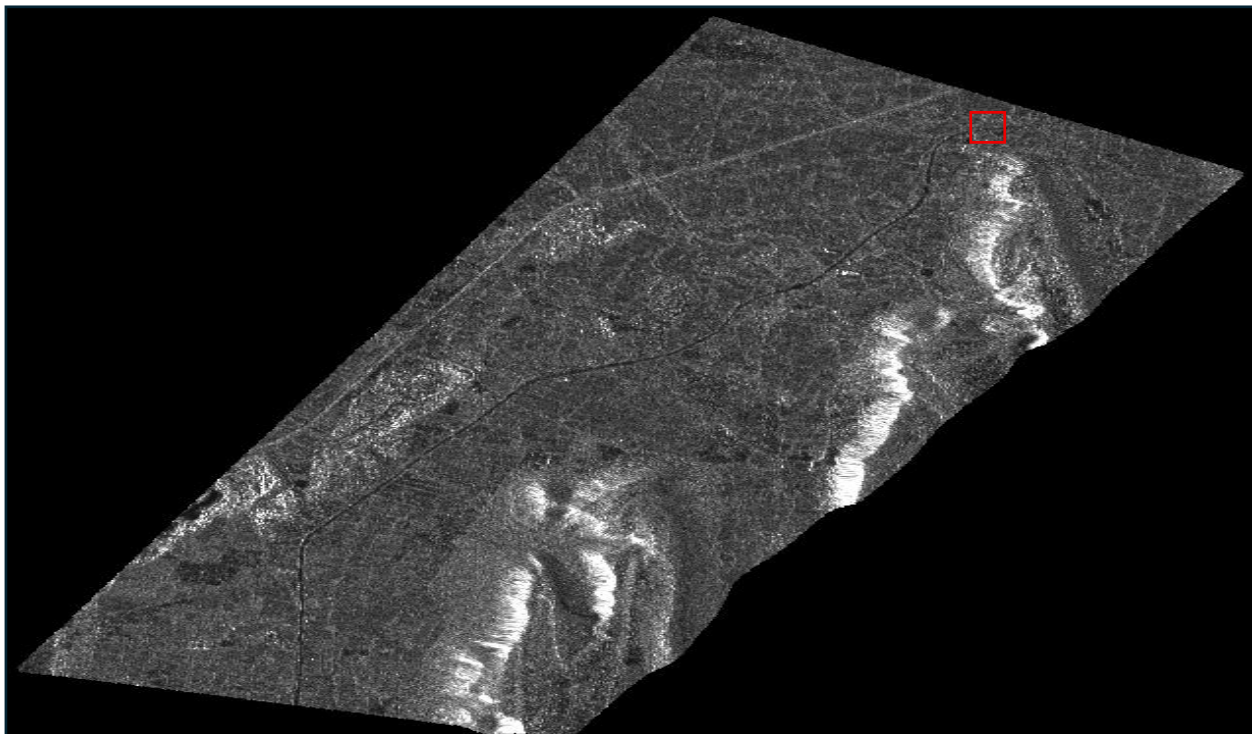
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization



3. Results of SAR based bridge inspection, including visualizations

Bridge Components

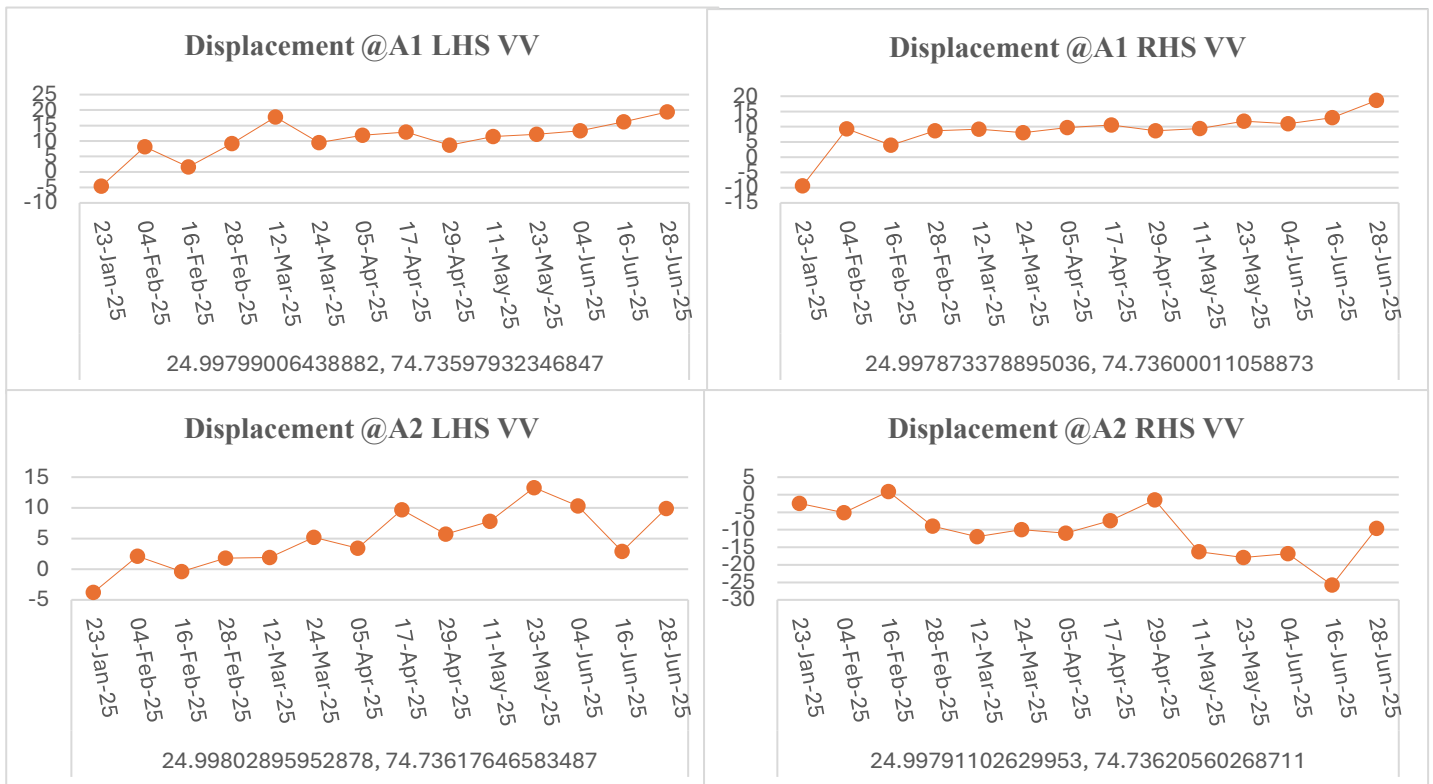
MNB – CH-920+200

Length – 21.16M

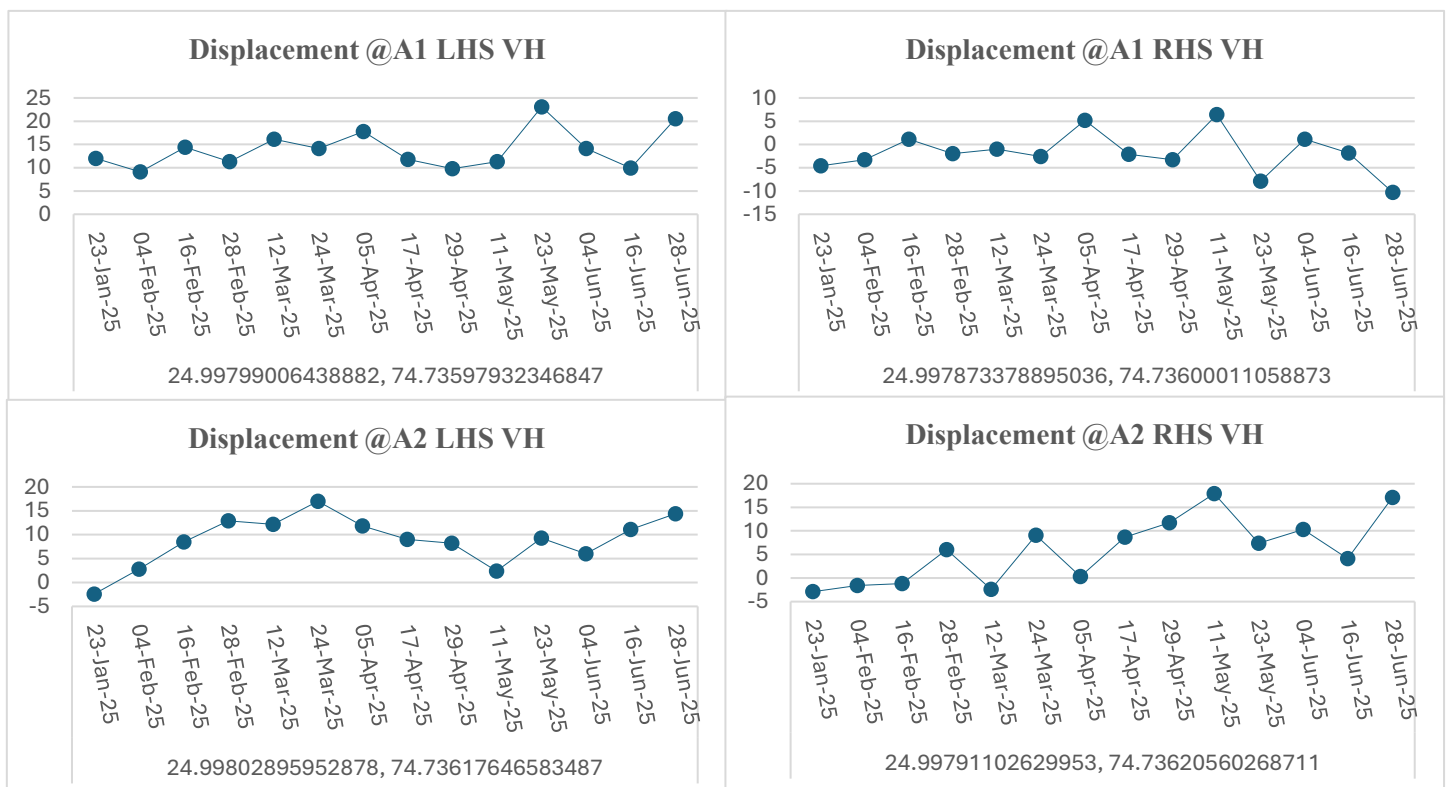


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



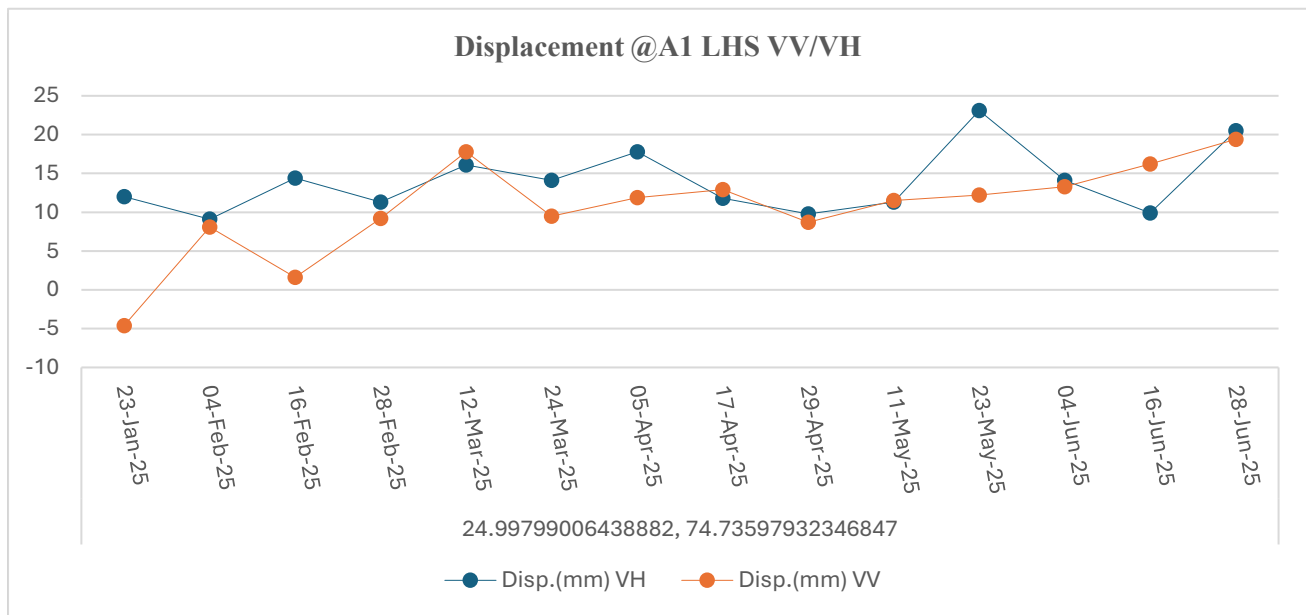
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.99799006438882, 74.73597932346847)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** values are predominantly positive, ranging approximately from +2 mm to +19 mm, indicating overall upward movement or uplift response at A1 LHS.
- **Horizontal Displacement (VH)** values remain consistently positive throughout the monitoring period, generally ranging between approximately +8 mm and +24 mm

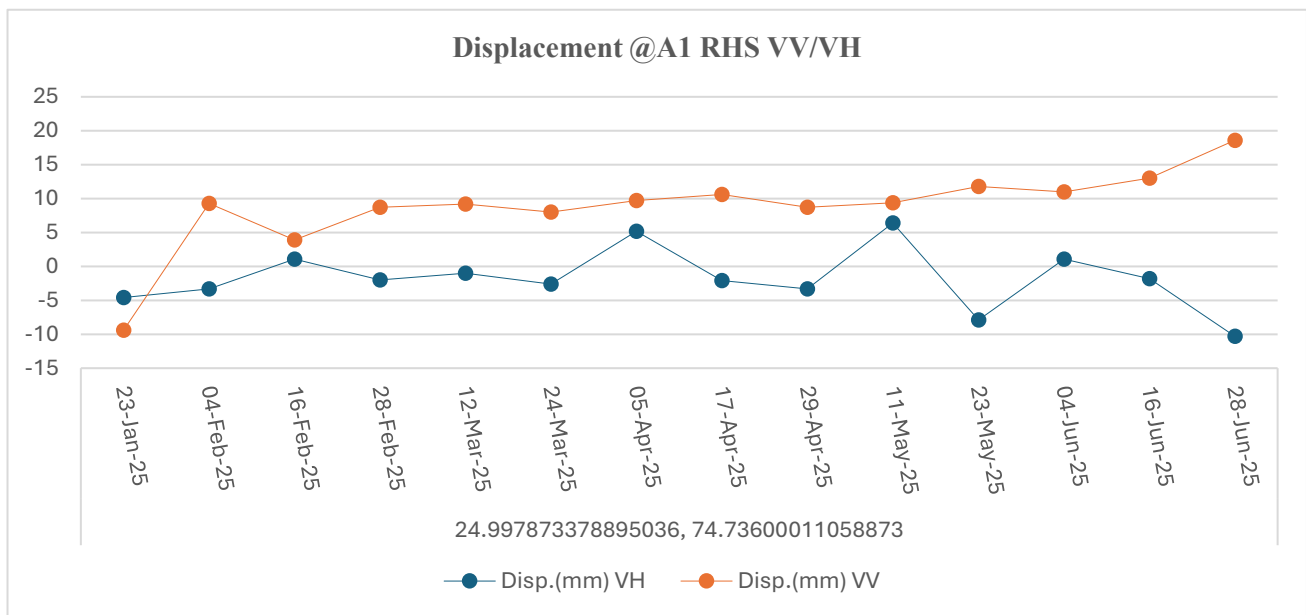
Critical Events & Anomalies:

- **March 2025:** Both VH and VV increase concurrently, indicating combined lateral and vertical response, potentially influenced by temperature rise or increased service loading.
- **May 2025:** VH reaches its highest recorded value while VV remains elevated, suggesting a peak structural response phase rather than isolated movement
- **June 2025:** VH drops again to -12 mm, while VV stabilizes near +5 mm, confirming continued lateral settlement toward the end of the monitoring period.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.997873378895036, 74.73600011058873)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** values exhibit bidirectional behaviour, fluctuating between approximately -10 mm and +6 mm, indicating alternating lateral movement directions at A1 RHS.
- **Vertical displacement (VV)** values remain predominantly positive throughout the monitoring period, ranging approximately from +3 mm to +18 mm, indicating consistent upward movement / uplift behaviour at A1 RHS.

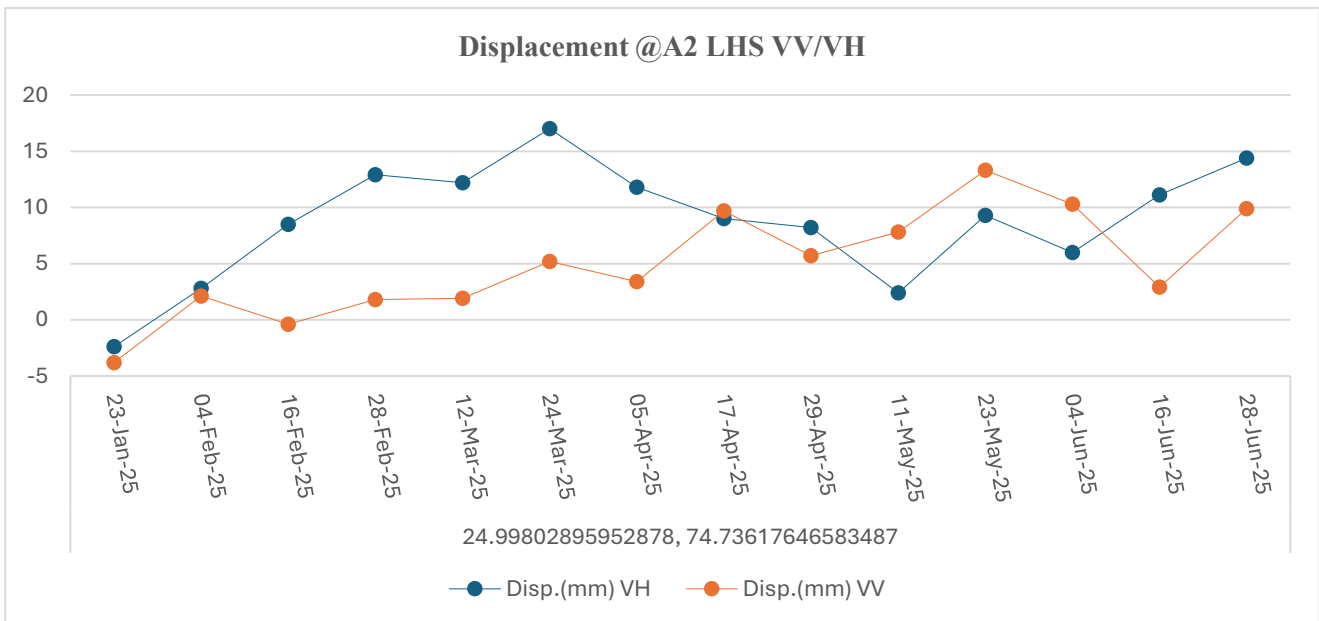
Critical Events & Anomalies:

- **May 2025:** A sudden reversal in VH direction is observed, transitioning to notable negative displacement while VV continues its upward trend. This indicates decoupled vertical and horizontal behaviour, not associated with structural distress.
- **June 2025:** VV reaches its highest magnitude, while VH remains within moderate limits, confirming dominant vertical response during this period.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.99802895952878, 74.73617646583487)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits moderate variability, ranging approximately from -3 mm to +17 mm, showing bidirectional lateral movement with a dominant positive trend.
- **Vertical Displacement (VV)** Vertical displacement values show predominantly positive behaviour, ranging approximately from -4 mm to +13 mm, indicating overall uplift-dominated vertical response at A2 LHS.

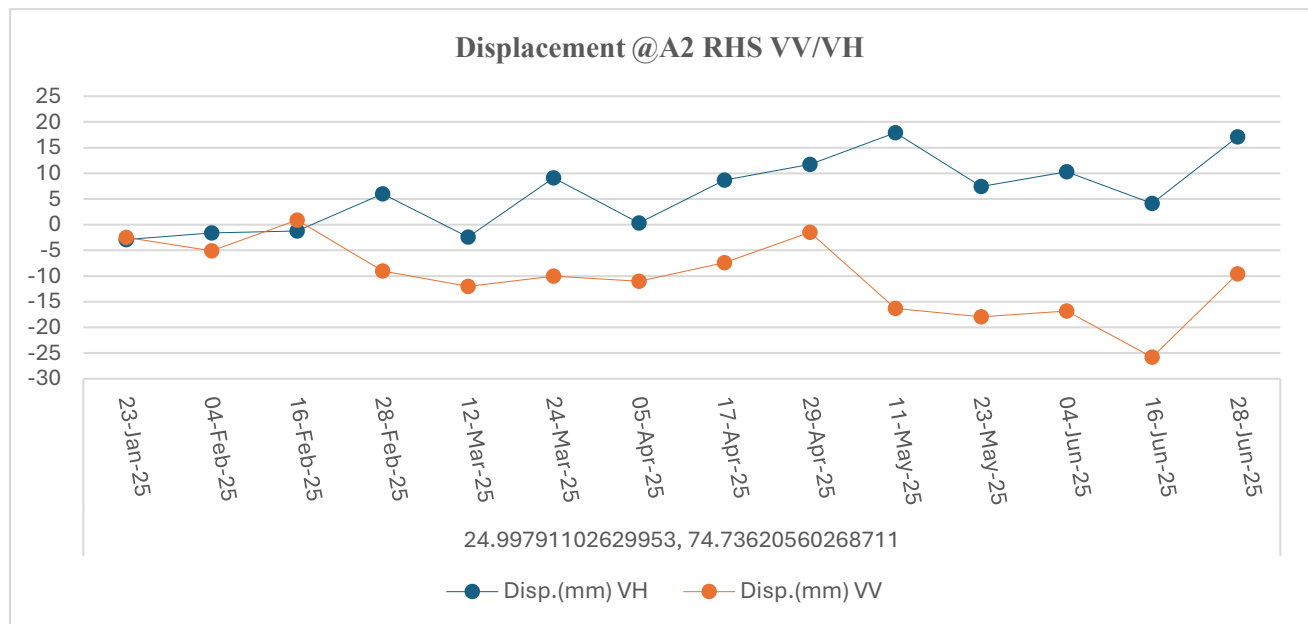
Critical Events & Anomalies:

- **March 2025:** A pronounced positive VH peak occurs while VV remains moderate and stable, suggesting a temporary lateral influence, likely driven by thermal effects or transient loading.
- **April-May 2025:** Both VH and VV show a simultaneous reduction, indicating a short-term relaxation phase rather than structural distress.
- **May 2025:** VV reaches its maximum uplift, while VH remains moderate, highlighting dominant vertical response during this period.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.99791102629953, 74.73620560268711)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacement values are predominantly negative, ranging approximately from -4 mm to -26 mm, indicating overall settlement-dominated vertical behaviour at A2 RHS.
- **Horizontal (VH) displacement** Horizontal displacement exhibits moderate to high variability, ranging approximately from -4 mm to $+18$ mm, with a clear predominance of positive lateral movement.

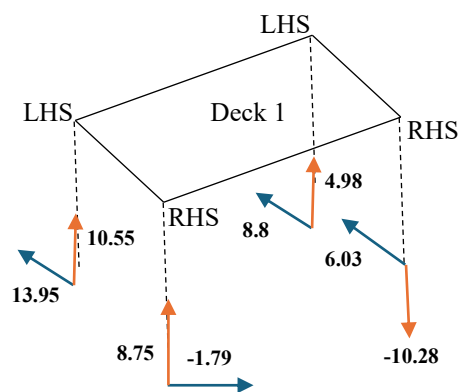
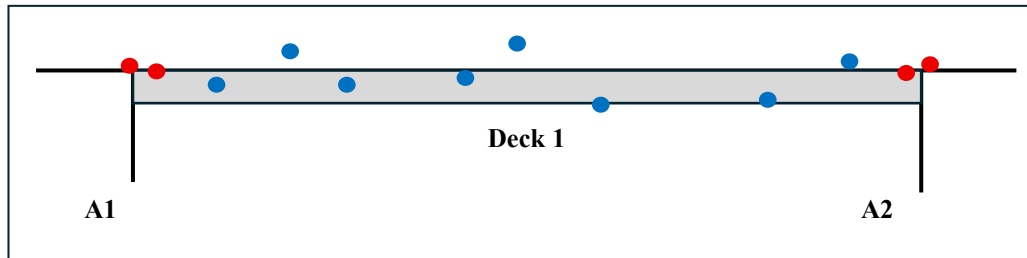
Critical Events & Anomalies:

- **May 2025:** A simultaneous VH peak and sharp VV drop is observed, indicating a combined lateral shift and vertical settlement event, likely driven by temperature variation, bearing response, or transient loading effects.
- **June 2025:** Partial vertical recovery combined with stable positive VH indicates elastic behaviour rather than progressive distress.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis

Vertical Displacement Behaviour:

- The bridge deck slab exhibits non-uniform vertical displacement across the LHS and RHS regions, indicating differential structural response. The maximum uplift is observed at the upper LHS deck region, reaching approximately +10.6 mm, representing the dominant vertical response in this configuration.

Horizontal Displacement Behaviour:

- Horizontal displacement shows clear directional variability, highlighting asymmetric and torsional behaviour of the deck slab. The RHS region exhibits moderate lateral displacement of approximately +6.0 mm, indicating secondary horizontal response compared to the LHS. The central lower deck region shows relatively minor lateral movement, suggesting proximity to a rotational or neutral axis.

CONCLUSION:

- The bridge deck slab demonstrates a distinctly non-symmetric deformation pattern, characterized by pronounced uplift and lateral movement on the LHS and localized settlement on the RHS. The presence of a near-neutral horizontal displacement zone near the deck centre suggests rotational behaviour rather than uniform translation. Overall, the displacement behaviour reflects asymmetric stiffness or loading conditions, reinforcing the importance of continued monitoring and comparative assessment across deck locations.